EXISTENCE OF INJECTIVE HULLS BY ZORN'S LEMMA

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-dualizing/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The existence of enough injective modules over a ring. Invoked at the start of the counted proof (source lines 703-705). Provides an injective module into which the given module embeds.
- The stability of the class of essential extensions under unions of totally ordered families. Invoked inside the counted proof (source lines 710-711). Shows the ordering by inclusion on essential extensions is inductive.
- The criterion for a maximal essential extension inside an injective module to be an injective hull. Invoked at the end of the counted proof (source lines 714-715). Supplies injectivity of the maximal element, which is the asserted hull.

Source reference. Original lemma lemma-essential-extensions-in-injective: sources/stacks-dualizing/pinned-original-source.tex, source lines 444–444. Navigation pointer only: the counted text refers to the Stacks item lemma-essential-extensions-in-injective, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-union-essential-extensions: sources/stacks-dualizing/pinned-original-source.tex, source lines 358–358. Navigation pointer only: the counted text refers to the Stacks item lemma-union-essential-extensions, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original reference more-algebra-section-injectives-modules: sources/stacks-dualizing/pinned-original-source.tex, source lines 459–459. Navigation pointer only: the counted text refers to the Stacks item more-algebra-section-injectives-modules,

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which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let R be a ring. Any R-module has an injective hull.*

Proof. Let M be an R-module. By More on Algebra, Section more-algebra-section-injectives-modules the category of R-modules has enough injectives. Choose an injection $M \to I$ with I an injective R-module. Consider the set $\mathcal{S}$ of submodules $M \subset E \subset I$ such that E is an essential extension of M. We order $\mathcal{S}$ by inclusion. If $\{E_\alpha\}$ is a totally ordered subset of $\mathcal{S}$, then $\bigcup E_\alpha$ is an essential extension of M too (Lemma lemma-union-essential-extensions). Thus we can apply Zorn's lemma and find a maximal element $E \in \mathcal{S}$. We claim $M \subset E$ is an injective hull, i.e., E is an injective R-module. This follows from Lemma lemma-essential-extensions-in-injective. $\square$